

The background of the slide is a dark gray. On the right side, there is a complex network of thin black lines connecting various colored dots (blue, red, green, purple, black). This network is partially enclosed by a thin, curved line that resembles a sphere or a dome. To the left of this network, there are many small, scattered black dots, similar to a star field or a sparse data set.

Ethics, Machine Ethics, Computational Ethics and the Law

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What is morality/ethics?

Ethics/Morality is about how we decide **what is right and wrong**, and how we evaluate the actions of ourselves and others.

In deciding what to do we can adopt **different perspectives**

1 Individual perspective

We can take our individual perspective, focusing on **our particular interests** (self-interest)

2 Moral perspective

We can be motivated by the belief that an action is right, regardless of how it affects our interest (morality/ethics)

What is morality/ethics?

Moral perspective

We can distinguish between



Positive (conventional) morality

Moral rules and **principles widely accepted in a society**. Used by members of a society to govern their lives. It can differ from society to society. **Can there be bad positive morality?**

Example: Conventional morality of Saudi Arabia forbids women from publicly contradicting their husbands or brothers, while Denmark's conventional morality allows this. People in the US would think it immoral to leave a restaurant without tipping a good waiter or bartender, while such behavior in many other societies is perfectly OK.



Critical morality

The morality that is **correct, rational, just** (e.g., since considers all individual and social interests at stake (harms to other, impacts on environment, etc.)

- It does not have its origin in social agreements
- It is untainted by mistaken beliefs, irrationality, or popular prejudices
- It can serve as the true standard or determining when conventional morality has got it right and when it has fallen into error.

We can criticise positive morality based on our critical morality: we may be right or wrong

Morality and social norms

Is morality a social phenomenon? we absorb from our society whatever standard is considered to be morally obligatory, and make it our own

- How do we learn social morality? How can an AI system learn social morality?

Is morality only matter of social learning and (basically) imitation?

- Can we develop a critical attitude toward the morality of our society? On what basis (reason? intuition?)

Do we want AI systems to learn social morality or develop a critical attitude toward it?



The Branches of Moral Philosophy

Ethics vs Metaethics

Ethics

It is concerned with determining **what is morally required**, how one ought to behave.

Questions:

- What are our fundamental moral duties? What makes right actions right?
- Do the ends always justify the means, or there are certain actions that should never be done under any circumstances?
- Is abortion right or wrong?
- Is compensation due for colonialism?

Metaethics

It is the study of the **nature, scope and meaning of moral judgement**

Questions:

- Can ethical judgments be true or false?
What is the difference between:
 - I prefer vegetables to meat
 - I ought to eat more vegetables to be more healthy
 - We ought to become vegetarians
- Do they correspond to some facts in the world?
- What facts make it true that we ought to become vegetarian? Or that we ought not to harm others?

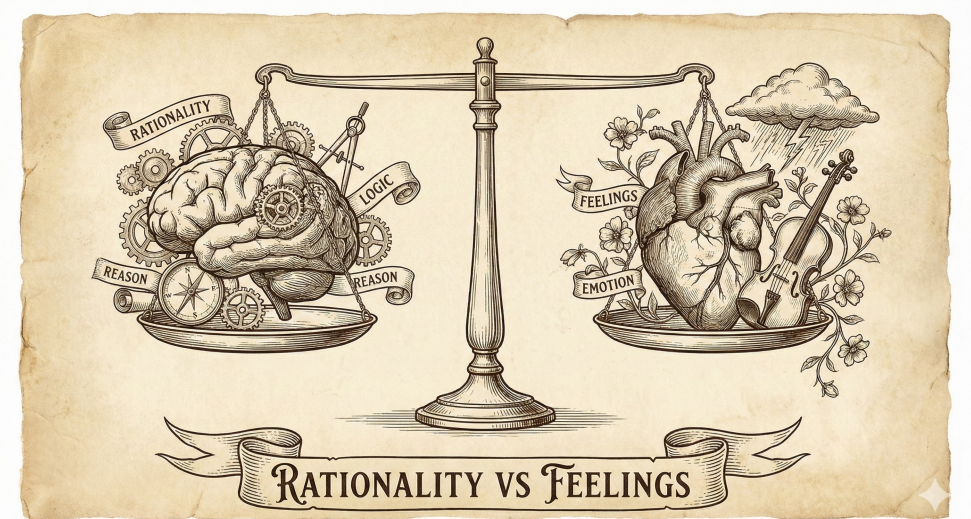
The Branches of Moral Philosophy

Ethics vs Metaethics

Rationality vs Feelings

Does ethic pertain to rationality of or feelings?

- **David Hume:** is not contrary to reason to prefer the destruction of the whole world to the scratching of my finger. Morality is a matter of sentiment (of impartial spectators)
- **Emmanuel Kant:** we can know what is moral through our reason
- **David Ross:** we can know what is more through our intuition





Absolutism vs Relativism

Absolutism

There is a single true ethics:
when 2 people express
incompatible ethical
judgement one of them
must be wrong

Relativism

Ethical judgements are always
**relative to particular
frameworks or attitudes**

A statement like "abortion is
morally permissible" or
"adultery is prohibited" or
"killing a willing person is
wrong" may be true under
some ethical framework and
wrong under some other

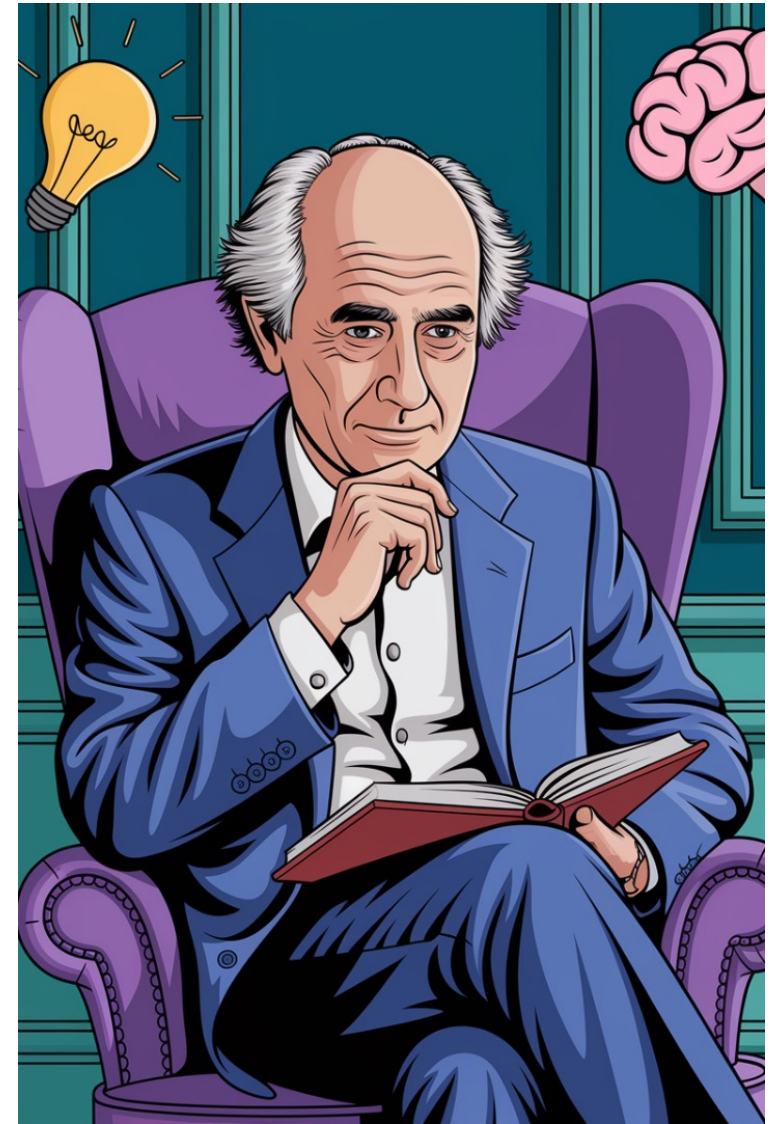
Analogy

As in mechanics, judgements are **relative to the frame of reference**
(a body may be moving relative to one frame and be stationary
relatively to another), so is in morality

Gilbert Harman

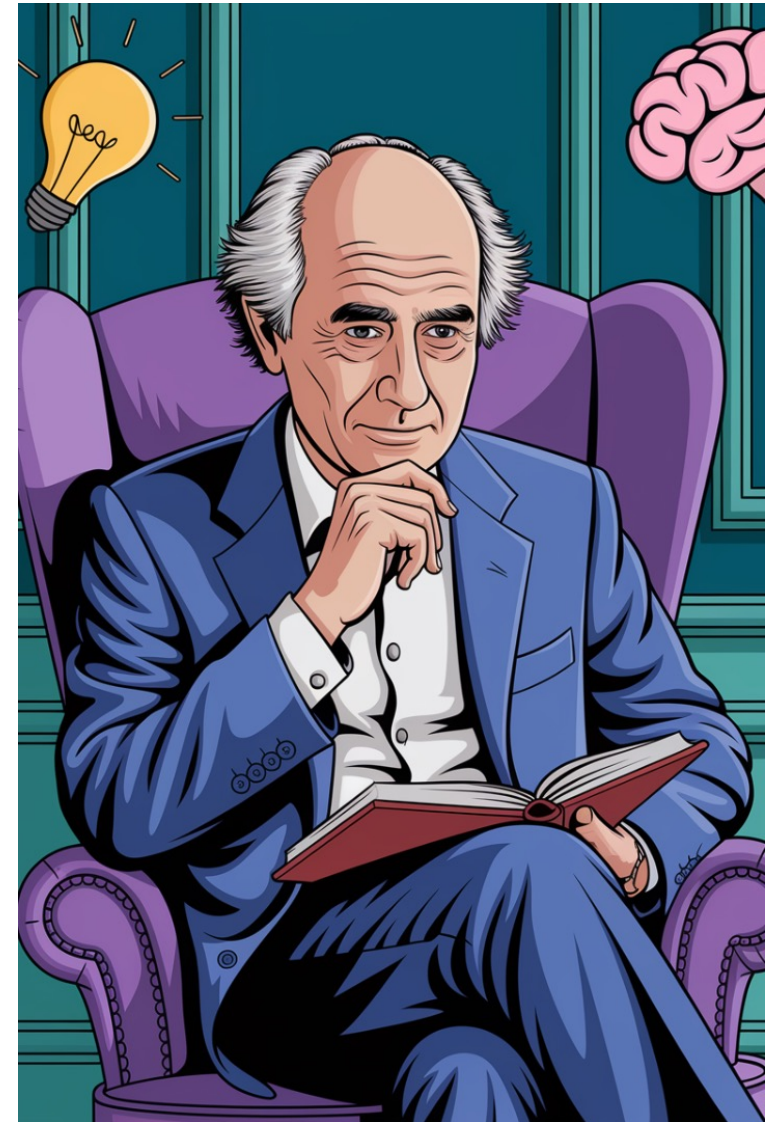
Gilbert Harman was a prominent American philosopher who made significant contributions to moral philosophy and epistemology.

Suppose Mabel takes Hitler's actions to be a great evil and also believes that Hitler's values were sufficiently perverse that they provided Hitler with no reason to refrain from acting as he acted [...]. Although she judges Hitler to be a great evil, she may find that she is no more able to judge that it was wrong of Hitler to have acted as he acted [...]."



Gilbert Harman

- This quote explores the challenging question of moral judgment
- What if we believe people's values and beliefs justify some actions in their mind?
- Harman suggests that even in cases of clear moral repugnance, like Hitler's actions, we may struggle to definitively condemn the person as "wrong" if his/her own ethical framework supported what he/she did.
- The complexities involved in making moral judgments, and the importance of trying to understand **someone's underlying beliefs and reasoning**, not just the surface-level actions.



Morality and disagreement

Areas of Disagreement

Morality is a place for widespread disagreement

- Abortion
- Migration
- Capital punishment
- Humanitarian wars
- ...

Potential Agreement

But there is something on which we may agree?

- It is wrong to kill innocent people?
- It is (usually) wrong to lie?
- It is (usually) wrong to harm people?

Pro-tanto and all-things-considered moral judgement

Defeasible Moral Prescriptions

Many moral prescription are not absolute but **defeasible**. They state general propositions that are susceptible of exceptions.

- We should not lie
- What if a lie would save a person's life?

Prima Facie Duty

An act is a **prima facie duty**: an act that is a moral reason in favor of doing it, but can be outweighed by other moral reasons

Robotic Agents and Defeasible Duties

Do we want a robotic agent to take its duties as defeasible?

David Ross's Example

David Ross: "If I have promised to meet a friend at a particular time for some trivial purpose, I should certainly think myself justified in breaking my engagement if by doing so I could prevent a serious accident or bring relief to the victims of one."

Morality and other normative systems



Law

- Does positive or critical morality include all laws enforced by the state? Does it include only such laws?
- Should all morally required behaviour be also legally required? Should all morality permitted behaviour also be legally permitted?



Religion

- Does critical morality include all and only what has been commanded by God?
- Did God command something because it was moral, or did anything become moral for having been commanded by God (rationalism vs voluntarism).
- Are atheists necessarily immoral or amoral? Is an atheistic society necessarily more immoral than a religious society?



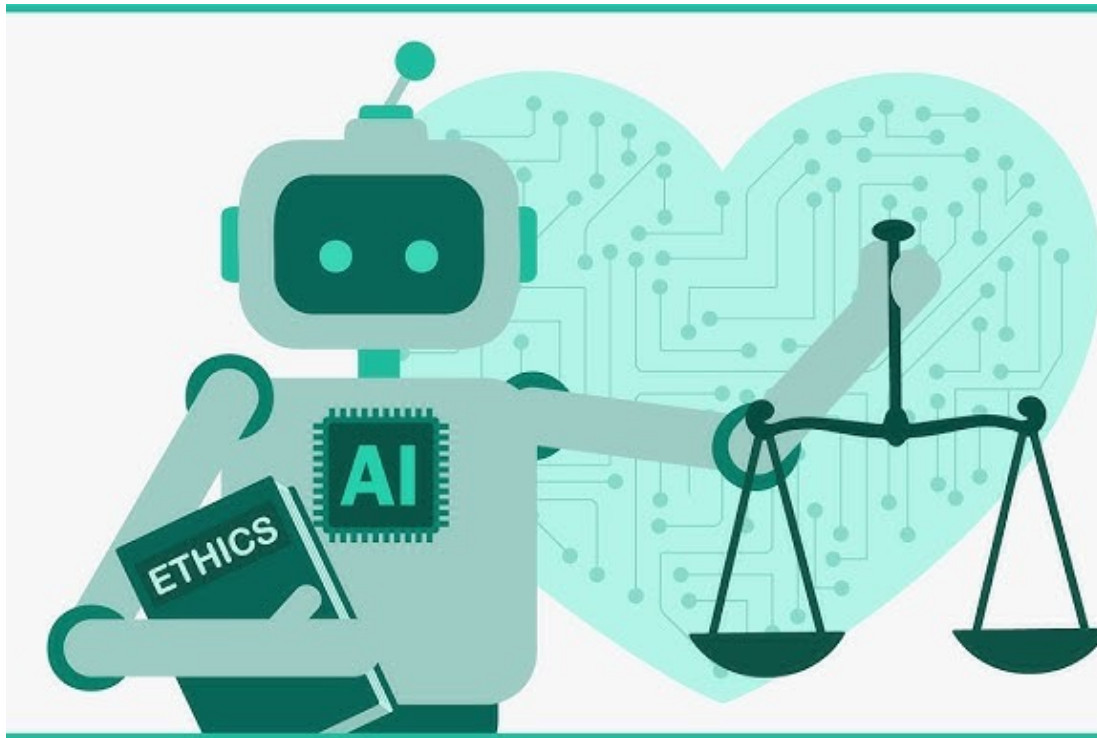
Tradition

- Source of moral guidance - how much weight should we give to **historical practices and norms**?



Self interest

- May morality and self interest collapse: should we do all and only what fits our personal interest (Gige's ring)
- Can morality be reduced to simply doing what's best for ourselves, or is there more to it?



Ethics and Ethics of AI

Ethics is about **how we decide what is right and wrong**, + **how we evaluate** the **actions** of ourselves and others (we may take different perspectives)

Ethics of AI refers to the broader **philosophical, social, and policy-related discussions** on how AI should be developed, deployed, and regulated in society.

Questions:

- Should AI be allowed to make life-and-death decisions?
- How do we ensure fairness, accountability, and transparency in AI systems?
- What ethical frameworks should guide AI governance?

Focus: Primarily concerns human responsibility in designing and using AI systems (e.g., Grok and non consensual sexual Deepfake)

Example: Debates on the ethical risks of AI in warfare or the use of AI for mass surveillance (e.g., Pegasus, Palantir ...)

What's the difference?

Ethics of AI



Ethical implication of AI use
(ee., regulation of privacy, AI biases)

Computational Ethics



Implementing ethical reasoning in AI systems
(ee., AI advising judges on ethical rulings)

Machine Ethics



Designing AI that makes autonomous
ethical decisions (ee., a robot deciding
a patient's request)

Law



Legal frameworks, regulations and rights
around AI (ee. GDPR, DSA, AI Act)

Ethics of AI

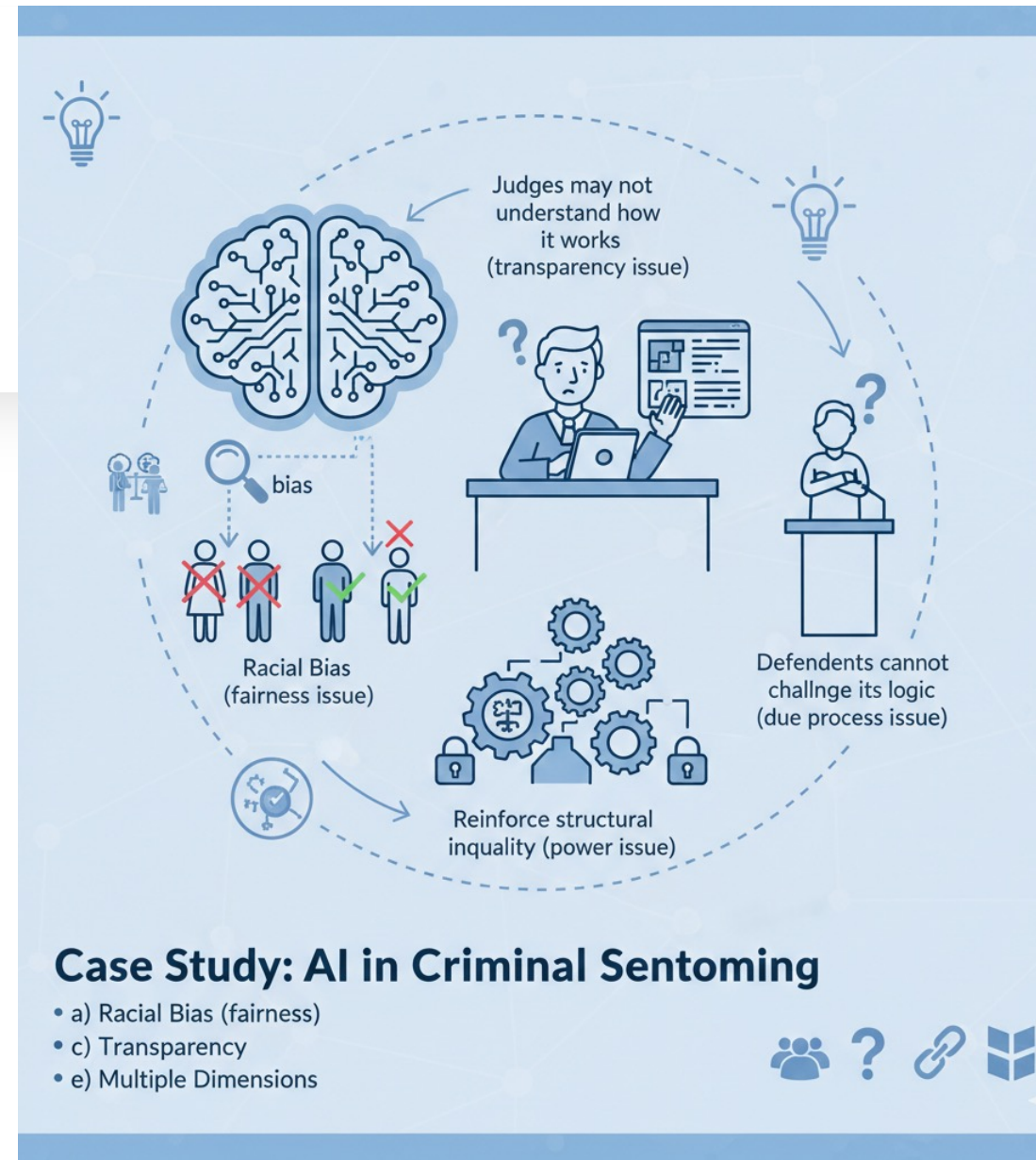
The study of, social, and political implications of designing, deploying, and governing artificial intelligence systems.

Key Questions:

- How much data collection is legitimate?
- Does AI treat individuals and groups equitably?
- Can we understand and explain how AI reaches its decisions?
- Who is responsible when AI causes harm?
-

Focus: not only on what AI can do, but on what it should do, under what constraints, and in whose interest.

Example: Consider an AI system used in criminal sentencing.



(AI) Computational Ethics

The study of how ethical reasoning can be modelled and implemented using computational methods, often for decision support in AI systems.

Key Questions:

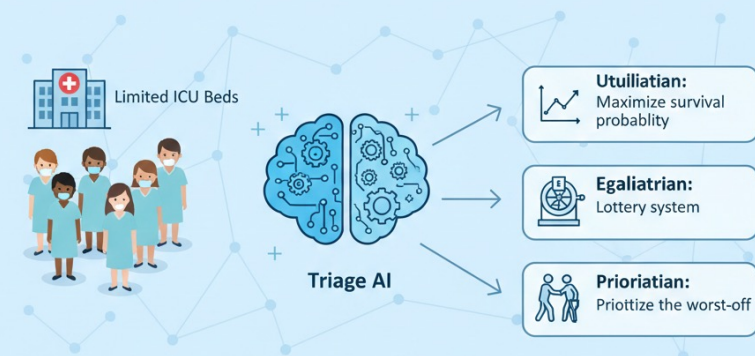
- How can we encode ethical principles (e.g., utilitarianism, deontology) into an algorithm?
- Can machine learning models detect ethical dilemmas and suggest moral actions?
- How do we formalize ethical reasoning in computational logic?

Focus: Developing algorithms and models that can analyze ethical situations and apply ethical rules.

Example: Medical triage AI during a pandemic.

Case Study: Medical Triage AI During a Pandemic

The system must allocate limited ICU beds.



The Computational Ethics Challenge:

How do we transform abstract moral theories into programmable decision criteria?

Machine Ethics

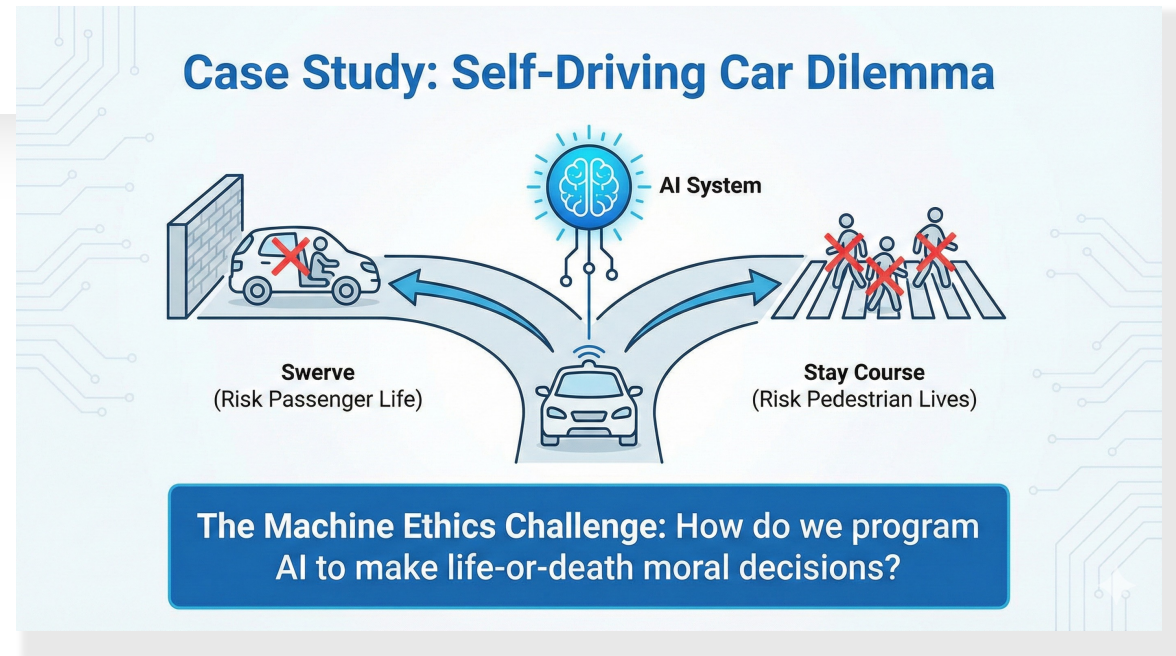
Focuses on making AI itself a moral agent, capable of ethical reasoning and decision-making.

Key Questions:

- Can machines have moral agency?
- How can we create autonomous systems that behave ethically?
- What moral frameworks should guide AI's decision-making?

Focus: The design of AI systems that can make independent ethical judgments, rather than relying on humans.

Example: A self-driving car in unavoidable accident scenario



AI and Law

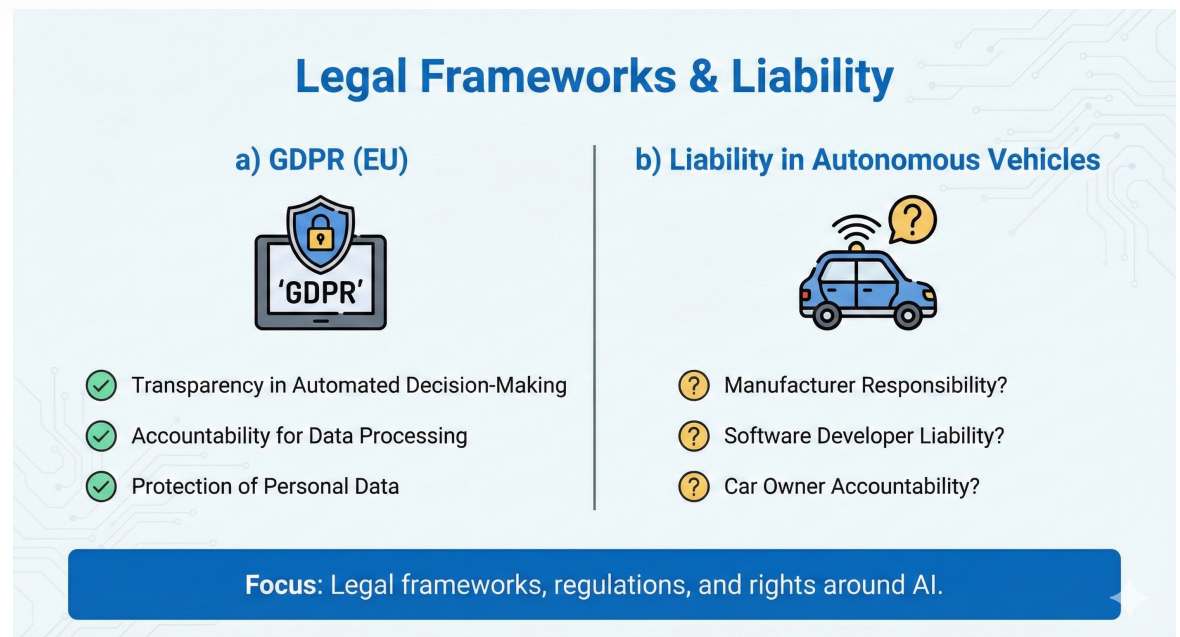
The legal frameworks and regulations that govern the development, deployment, and use of AI systems. It encompasses the rules, rights, and obligations surrounding AI technologies and addresses the legal challenges posed by AI in society.

Key Questions:

- How should AI systems be regulated to protect privacy, security, and fairness?
- What legal liability exists when AI systems cause harm (e.g., accidents, discrimination)?
- How can intellectual property laws apply to AI-generated content? ...

Focus: establishing laws, regulations, and standards balancing innovation with protection of human rights and societal welfare.

Example: GDPR, AI Act, ...





Consequentialism

Focuses on the outcomes or consequences of actions.



Utilitarianism

Greatest good for the greatest number of people.



Deontology

Emphasizes moral duties, rules, and obligations.



Virtue Ethics

Focuses on the character and virtues of the moral agent.



Which Ethical Theory?



Questions?

